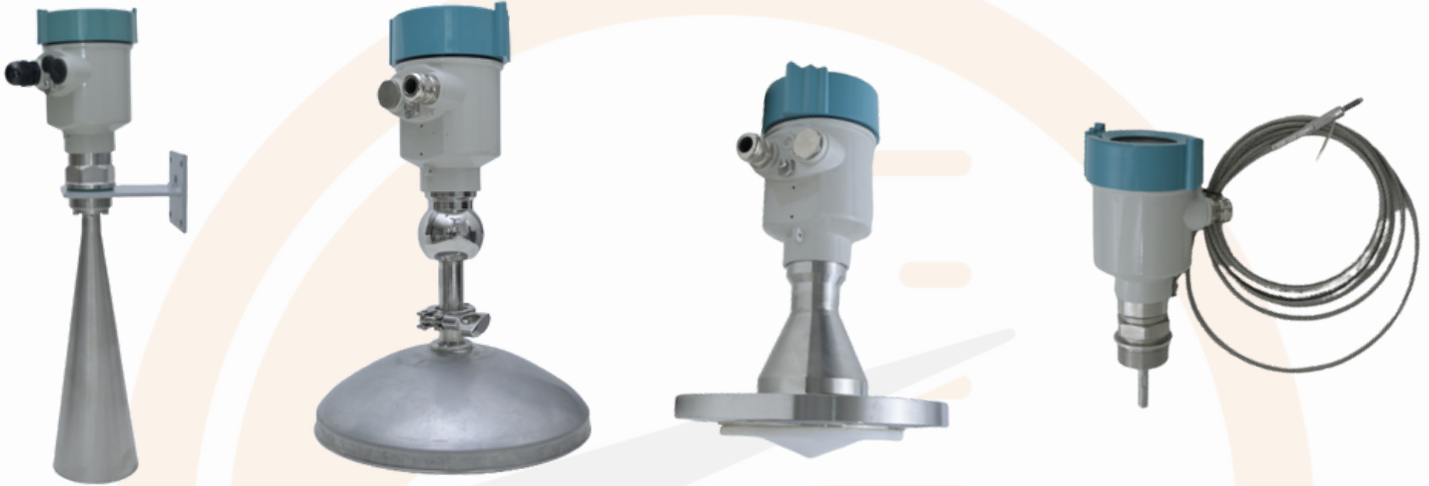


BROTEK Radar level meter antenna by narrow microwave pulse, the pulse propagation in space at the speed of light, meet the measured medium surface, the part of the energy to be reflected back, by the same antenna. Emission pulses and the time interval of the received pulse and the antenna to the measured medium surface is proportional to the distance. Due to the electromagnetic wave propagation and very high speed, pulse and receiving pulse time interval is very small (nanosecond) it is difficult to confirm, 26G radar level meter adopts a special demodulation technology, can accurately identify the transmitted pulse and pulse interval, thus further calculate the antenna to the measured medium surface distance. It is a new generation radar level meter with full independent property rights developed by Brotek after many years of hard work.

ATVUK-PMR-26-XXXX

ATVUK-PMR-G-XXXX



Technical Details

Range: 10m/ 20m/ 30m/ 70m
Connection: Threaded/ Flanged
Power Supply: 24VDC
Output: 4-20mA/HART/RS485
Measuring Accuracy: $\pm 3\text{mm}$ / $\pm 5\text{mm}$ / $\pm 10\text{mm}$ /
 $\pm 15\text{mm}$
Temperature: -40°C to 130°C / -40°C to 250°C
Pressure: may vary on model
Frequency Range: 6GHz/ 26GHz
Beam Angle: 3° / 8° / 12°
Display: Inbuild
Antenna Type: Horn/ Dish/ Capsule
Antenna Material: PTFE/ Stainless Steel
Enclosure: Die-cast Aluminium
Protection Rating: IP67

Features

- Versatility for various applications
- High accuracy and repeatability
- Almost no corrosion, bubble effect, almost not affected by water vapor in atmosphere, the temperature and pressure changes.
- High signal to noise ratio.
- High frequency, measurement of solid and low dielectric constant of the best choice.
- A shorter wavelength, the reflection of solid surface inclination is better.
- Easy to onsite calibration
- Enhanced safety (no direct contact with material)

Applications

Radar level transmitters are commonly used in a variety of industrial applications where non-contact, reliable and highly accurate level measurement is required. Some examples are Chemical and petrochemical processing plants and refineries, Food and beverage processing, Water treatment plants, Pharmaceutical manufacturing, Wastewater treatment plants, Mining and material handling.



BROTEK FMCW radar (Frequency-Modulated Continuous Wave radar) adopt the millimeter wave band with higher frequency. In pulse radar, however, the runtime must be measured directly. Whereas, FMCW radar are measured the differences in phase or frequency between the actually transmitted and the received signal instead. With extremely narrow beam and penetration ability, it can better adapt to ultra complex working conditions without weakening the measurement performance. Based on the self-developed CMOS millimeter wave RF chip, a more compact RF architecture, a higher signal-to-noise ratio, and a smaller blind zone are realized. It is a new generation radar level meter with full independent property rights developed by Brotek after many years of hard work.

ATVUK-FMCW-XXXX



Technical Details

Range: 10m/ 20m/ 60m/ 120m
Connection: Threaded/ Flanged
Power Supply: 24VDC
Output: 4-20mA/HART/RS485
Measuring Accuracy: $\pm 1\text{mm}$ / $\pm 5\text{mm}$ / $\pm 10\text{mm}$
Temperature: may vary on model
Pressure: may vary on model
Frequency Range: 80GHz/ 120GHz
Beam Angle: 3° / 8°
Display: Inbuild
Antenna Type: Capsule
Antenna Material: PTFE
Enclosure: ABS plastic/ Die-cast Aluminium/
Stainless Steel
Protection Rating: IP67

Features

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- High accuracy and repeatability
- Almost no corrosion, bubble effect, almost not affected by water vapor in atmosphere, the temperature and pressure changes.
- High signal to noise ratio.
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Applications

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